

LECTURE 12: LINEAR INSTRUMENTAL VARIABLES

Guido Imbens – Stanford University

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OUTLINE

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2. Linear Instrumental Variables with Constant Coefficients
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7. Limited Information Maximum Likelihood
8. LIML vs TSLS: Does it matter?

ENDOGENEITY

- Consider a regression model when one of the regressors is **endogenous**,

$$\widehat{\log - \text{wage}}_i = \beta_0 + \beta_1 \times \text{educ}_i + \varepsilon_i$$

so that we **cannot** think of the regressor as (conditionally) randomly assigned. It may be correlated with the residual, ε_i .

- Let us look at the regression of log wages on years of education. Using the Angrist-Krueger (1991) data we get, using least squares,

$$\widehat{\log - \text{wage}}_i = \hat{\beta}_0 + 0.0709 \times \text{educ}_i \text{ (s.e. 0.0003)}$$

- Standard error is very small, because the sample size is large ($N = 329,509$)
One component of ε_i may be **ability**, and that may be correlated with **education**.

LINEAR INSTRUMENTAL VARIABLES WITH CONSTANT COEFFICIENTS

- We are interested in the causal effect of years of education on log earnings.
- Let Y_i be the outcome for unit i , log earnings, let X_i the regressor, years of education.
- A linear model is postulated for the relation between the outcome and the regressor:

$$Y_i = \beta_0 + \beta_1 \times X_i + \varepsilon_i.$$

Previously we discussed the best linear prediction interpretation of a regression function, that simply defines β_1 in terms of the correlation between X_i and Y_i .

- However, here we are after something different. a **structural/behavioral/causal** one, with β_1 capturing a **causal** effect, not simply a **descriptive** relationship.
- Think of each individual/unit having a causal relationship:

$$Y_i(x) = \beta_0 + \beta_1(x) + \varepsilon_i.$$

This describes for unit i what you would see if you **changed** x .

ENDOGENEITY

- There is concern that the regressor X_i is **endogenous**, that is, that X_i is correlated with ε_i .
- Individuals may choose X_i depending on the function $Y_i(\cdot)$, or prices may be set depending on the demand function.
- Suppose that we are confident that a second variable, the instrument Z_i is **uncorrelated** with the unobserved component but **correlated** with the endogenous regressor.
- The solution is to use Z_i as an **instrument** for X_i .

TWO-STAGE-LEAST-SQUARES

- In **Two-Stage-Least-Squares** there are two stages:
 - In the first stage we first estimate a linear regression of the endogenous regressor on the instrument by least squares. Let the estimated regression function be

$$\hat{X}_i = \hat{\pi}_0 + \hat{\pi}_1 \cdot Z_i.$$

- In the second stage we regress the **outcome** Y_i on the **predicted value of the endogenous regressor** $\hat{X}_i$ (as opposed to the **actual value of the endogenous regressor**, X_i), using least squares:

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1^{\text{TOLS}} \cdot \hat{X}_i.$$

- This leads to the two-stage-least-squares estimator

$$\hat{\beta}_1^{\text{TOLS}} \equiv \frac{\text{cov}(Y_i, Z_i)}{\text{cov}(X_i, Z_i)}$$

INDIRECT LEAST SQUARES

- Alternatively, with a single instrument we can estimate the two **reduced form regressions** (as opposed to the **structural form regression** of Y_i on X_i)

$$Y_i = \gamma_0 + \gamma_1 \cdot Z_i + \eta_i,$$

$$X_i = \pi_0 + \pi_1 \cdot Z_i + \nu_i.$$

- We estimate both by **least squares**.
- Then we estimate β_1 as the ratio

$$\hat{\beta}_1^{\text{ILS}} = \hat{\gamma}_1 / \hat{\pi}_1.$$

This is known as **Indirect Least Squares** (ILS).

EQUALITY OF TSLS AND ILS

- If there is a **single instrument** and **single endogenous regressor**, we end up in both cases with the ratio of the sample covariance of Y and Z to the sample covariance of W and Z .

$$\hat{\beta}_1 = \hat{\beta}_1^{\text{TSLS}} = \hat{\beta}_1^{\text{ILS}} = \frac{\frac{1}{N} \sum_{i=1}^N (Y_i - \bar{Y}) \cdot (Z_i - \bar{Z})}{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X}) \cdot (Z_i - \bar{Z})}.$$

- With **multiple instruments** TSLS and ILS generally differ. (There are multiple ratios that can be used in the ILS derivation.)

INFERENCE WITH INSTRUMENTAL VARIABLES

- Write the model as

$$Y_i = X_i^\top \beta + \varepsilon_i$$

- Full set of regressors $X_i = (X_{i1}, X_{i2})$ may include endogenous regressors X_{i1} and exogenous regressors X_{i2} (with the latter typically including the intercept).
- Instruments $Z_i = (W_i, X_{i2})$ may include exo regressors X_{i2} and excluded instruments W_i .
- Define the projection of the regressors on the set of instruments and TSLS estimator

$$\hat{\mathbf{X}} \equiv \mathbf{Z} \left(\mathbf{Z}^\top \mathbf{Z} \right)^{-1} \left(\mathbf{Z}^\top \mathbf{X} \right) \quad \hat{\beta}^{\text{TSLS}} \equiv \left(\hat{\mathbf{X}}^\top \hat{\mathbf{X}} \right)^{-1} \left(\hat{\mathbf{X}}^\top \mathbf{Y} \right) = \left(\mathbf{Z}^\top \mathbf{X} \right)^{-1} \left(\mathbf{Z}^\top \mathbf{Y} \right)$$

- Define $\hat{\varepsilon}_i = Y_i - \hat{X}_i^\top \hat{\beta}^{\text{TSLS}}$
- Then, under homoskedasticity

$$\hat{\beta}^{\text{TSLS}} - \beta \approx \mathcal{N} \left(0, \sigma^2 (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \right)$$

without homoskedasticity you can use GMM machinery to get the asymptotic variance.

BACK TO ANGRIST-KRUEGER (1991) APPLICATION

- In the Angrist and Krueger case the instrument is quarter of birth. Here I use an indicator for quarter of birth equal to the first quarter as the single binary instrument, leading to:

$$\widehat{\log - \text{wage}}_j = \hat{\beta}_0 + 0.1020 \times \text{educ}_j \text{ (s.e. 0.0239)}$$

- Standard error much higher for IV than for OLS

COMPARING INSTRUMENTAL VARIABLES ESTIMATES AND LEAST SQUARES ESTIMATES

- What if ε_i is **indep** of X_i ? Both TSLS and OLS are consistent, but OLS is efficient.
- We can compare TSLS and OLS as a **test** of the hypothesis that X_i is exogenous.
- This is known as a Hausman test (Hausman, 1978):

$$\hat{\beta}^{\text{TSLS}} - \hat{\beta}^{\text{OLS}} \approx \mathcal{N}\left(0, \hat{V}_{\text{TSLS}} - \hat{V}_{\text{OLS}}\right)$$

This relies on the fact that under the null hypothesis OLS is efficient, and the TSLS estimator is consistent but not efficient. implying

$$\mathbb{V}(\hat{\beta}^{\text{TSLS}} - \hat{\beta}^{\text{OLS}}) = \mathbb{V}(\hat{\beta}^{\text{TSLS}}) - \mathbb{V}(\hat{\beta}^{\text{OLS}})$$

- For the AK data:

$$T - stat = \frac{\hat{\beta}^{\text{TSLS}} - \hat{\beta}^{\text{OLS}}}{\sqrt{\hat{V}_{\text{TSLS}} - \hat{V}_{\text{OLS}}}} = -1.3$$

MULTIPLE INSTRUMENTS

- With multiple instruments (more instruments than endogenous regressors) there are different estimators. K regressors, M instruments, $K < M$.
- ILS is possible using one instrument at a time.
- Both Z_{1i} and Z_{2i} are uncorrelated with $Y_i - \beta_0 - \beta_1 \cdot X_i$. Then

$$\hat{\beta}_1^{\text{ILS}} = \frac{\frac{1}{N} \sum_{i=1}^N (Y_i - \bar{Y})(Z_{1i} - \bar{Z}_1)}{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})(Z_{1i} - \bar{Z}_1)}.$$

versus

$$\hat{\beta}_2^{\text{ILS}} = \frac{\frac{1}{N} \sum_{i=1}^N (Y_i - \bar{Y})(Z_{2i} - \bar{Z}_2)}{\frac{1}{N} \sum_{i=1}^N (X_i - \bar{X})(Z_{2i} - \bar{Z}_2)}.$$

are both valid.

- Alternatively, we can combine them in different ways. The two leading estimators are **TSLS** and **Limited Information Maximum Likelihood** (LIML).

TWO STAGE LEAST SQUARES

- The TSLS estimator is essentially the same as before.
- First calculate

$$\hat{\mathbf{X}} = \mathbf{Z}\hat{\pi} = \mathbf{Z} \left(\mathbf{Z}^{\top} \mathbf{Z} \right)^{-1} \left(\mathbf{Z}^{\top} \mathbf{X} \right) \quad \text{where } \hat{\pi} = \left(\mathbf{Z}^{\top} \mathbf{Z} \right)^{-1} \left(\mathbf{Z}^{\top} \mathbf{X} \right)$$

- Then

$$\hat{\beta}^{\text{TSLS}} = \left(\hat{\mathbf{X}}^{\top} \hat{\mathbf{X}} \right)^{-1} \left(\hat{\mathbf{X}}^{\top} \mathbf{Y} \right)$$

- Note that $\mathbf{Z}^{\top} \mathbf{X}$ is **not invertible** (is it not even a square matrix).

LIMITED INFORMATION MAXIMUM LIKELIHOOD

- **Limited Information Maximum Likelihood** (LILM) estimates all parameters (both π and β) in one step, by maximizing the log likelihood function based on normality:

$$L(\beta, \pi, \Omega) = \sum_{i=1}^N \left(-\frac{1}{2} \ln |\Omega| - \frac{1}{2} \begin{pmatrix} y_i - \beta z_i^\top \pi \\ x_i - z_i^\top \pi \end{pmatrix}' \Omega^{-1} \begin{pmatrix} y_i - \beta z_i^\top \pi \\ x_i - z_i^\top \pi \end{pmatrix} \right).$$

- The asymptotic distribution of the LIML and TSLS estimator are the **same**, without relying on normality, but relying on the model being correctly specified.

$$\hat{\beta}^{\text{TSLS}} - \beta \approx \hat{\beta}^{\text{LIML}} - \beta \approx$$

$$\mathcal{N} \left(0, \left(\sum_{i=1}^N \hat{x}_i \hat{x}_i^\top \right)^{-1} \left(\sum_{i=1}^N \hat{\varepsilon}_i^2 \hat{x}_i \hat{x}_i^\top \right) \left(\sum_{i=1}^N \hat{x}_i \hat{x}_i^\top \right)^{-1} \right)$$

K-CLASS ESTIMATORS

- For some k , regress Y_i on $(1 - k)X_i + k\hat{X}_i$

$$Y_i = ((1 - k)X_i + k\hat{X}_i)^\top \beta + \varepsilon_i$$

- Three important cases:
 - If $k = 0$ this is ols.
 - If $k = 1$, this is tsls
 - if k is the smallest root of $|\mathbf{W}_1 - \lambda \mathbf{W}| = 0$, this is liml
 - ★ Here $\mathbf{W}_1$ is the cross product of the residuals from regressing Y_i and the endogenous regressors on the exogenous regressors in the main equation, and $\mathbf{W}$ is the cross product of the residuals from regressing Y_i and the endogenous regressors on the all exogenous regressors (including the instruments)

LIML VS TSLS: DOES IT MATTER?

- Angrist-Krueger quarter-of-birth data.
- **Three** instruments (quarter of birth equal to 1, 2, 3, with quarter 4 left out).

TSLS : 0.1026 (0.0195), LIML : 0.1035 (0.0201)

- **Not very different.** This was the traditional view, and TSLS was more common because it was slightly easier to implement.
- **Warning:** The choice between TSLS and LIML does matter with **lots** of weak instruments, to be discussed next week.

SIMULTANEOUS EQUATIONS MODELS

- 1960s simultaneous equations models:

$$\mathbf{Y}\Gamma + \mathbf{X}\mathbf{B} + \mathbf{U} = 0$$

- $\mathbf{Y}$ is $N \times M$ matrix of endogenous regressors, $\mathbf{X}$ is $N \times K$ matrix of exogenous regressors. $\mathbf{U}$ is $N \times M$ matrix of error terms / residuals.
- Γ is a $M \times M$ matrix of parameters, $\mathbf{B}$ is a $K \times M$ matrix of parameters.
- Γ and $\mathbf{B}$ **must** have some restrictions to be identified.
- **caveat**: rare to see these models with more $M > 2$. But for while this was a big literature.

SIMULTANEOUS EQUATIONS MODELS

- Example I: Γ is the identity matrix. Then we just have M linear regressions with exogenous regressors.

SIMULTANEOUS EQUATIONS MODELS

- Example II

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad X_i = \pi_0 + \pi_1 Z_i + \eta_i$$

- How does this fit it?

$$\mathbf{Y} = \begin{pmatrix} Y_1 & X_1 \\ Y_2 & X_2 \\ \vdots & \vdots \\ Y_N & X_N \end{pmatrix} \quad \mathbf{X} = \begin{pmatrix} 1 & Z_1 \\ 1 & Z_2 \\ \vdots & \vdots \\ 1 & Z_N \end{pmatrix}$$

$$\Gamma = \begin{pmatrix} 1 & \beta_1 \\ 0 & 1 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} \beta_0 & 0 \\ \pi_0 & \pi_1 \end{pmatrix}$$

SIMULTANEOUS EQUATIONS MODELS: SUPPLY AND DEMAND

- Traditional supply and demand model:

$$\text{(demand)} \quad Q_t^D(p) = \alpha_0 + \alpha_1 p + \alpha_2 D_t + \varepsilon_t$$

(with demand shifter D_t)

$$\text{(supply)} \quad Q_t^S(p) = \beta_0 + \beta_1 p + \beta_2 S_t + \eta_t$$

(with supply shifter S_t)

$$\text{(equilibrium)} \quad Q_t^D(p_t^E) = Q_t^S(p_t^E)$$

- So we can write the model in terms of equilibrium quantities

$$Q_t^E = \alpha_0 + \alpha_1 P_t^E + \alpha_2 D_t + \varepsilon_t$$

$$Q_t^S = \beta_0 + \beta_1 P_t^E + \beta_2 S_t + \eta_t$$

SIMULTANEOUS EQUATIONS MODELS: SUPPLY AND DEMAND

- Equilibrium price (note: $\beta_1 < 0$, $\alpha_1 > 0$):

$$\alpha_0 + \alpha_1 P_t^E + \alpha_2 D_t + \varepsilon_t = \beta_0 + \beta_1 P_t^E + \beta_2 S_t + \eta_t$$

This implies the reduced form

$$P_t^E = \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1} + \frac{\beta_2}{\alpha_1 - \beta_1} S_t - \frac{\alpha_2}{\alpha_1 - \beta_1} D_t + \frac{\eta_t - \varepsilon_t}{\alpha_1 - \beta_1} = \pi_{00} + \pi_{01} S_t + \pi_{02} D_t + \kappa_t$$

$$Q_t^E = \frac{\alpha_1 \beta_0 - \alpha_0 \beta_1}{\alpha_1 - \beta_1} + \frac{\alpha_1 \beta_2}{\alpha_1 - \beta_1} S_t + \frac{\alpha_2 \beta_1}{\alpha_1 - \beta_1} D_t + \frac{\alpha_1 \eta_t - \beta_1 \varepsilon_t}{\alpha_1 - \beta_1} = \pi_{10} + \pi_{11} S_t + \pi_{12} D_t + \nu_t$$

- Can we infer $(\alpha_0, \alpha_1, \alpha_2, \beta_0, \beta_1, \beta_2)$ from $(\pi_{00}, \pi_{01}, \pi_{02}, \pi_{10}, \pi_{11}, \pi_{12})$?
- yes (subject to some regularity conditions, e.g., $\alpha_2 \neq 0$, $\beta_2 \neq 0$).

SIMULTANEOUS EQUATIONS MODELS: SUPPLY AND DEMAND

- In the general simultaneous equations form:

$$\mathbf{Y} = \begin{pmatrix} Q_1^E & P_1^E \\ Q_2^E & P_2^E \\ \vdots & \vdots \\ Q_N^E & P_N^E \end{pmatrix} \quad \mathbf{X} = \begin{pmatrix} 1 & S_1 & D_1 \\ 1 & S_2 & D_2 \\ \vdots & \vdots & \vdots \\ 1 & S_N & D_N \end{pmatrix}$$

$$\Gamma = \begin{pmatrix} 1 & 1 \\ \alpha_1 & \beta_1 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} \alpha_0 & \beta_0 \\ 0 & \beta_2 \\ \alpha_2 & 0 \end{pmatrix}$$

SIMULTANEOUS EQUATIONS MODELS

- Reduced form:

$$\mathbf{Y} = \mathbf{X}\Pi + \mathbf{E} = -\mathbf{XB}\Gamma^{-1} + \mathbf{U}\Gamma^{-1} \quad \Pi = -\mathbf{B}\Gamma^{-1}$$

- What can we say about identification for the general case?
- focus on the first equation, with Γ_i (B_i) is i -th column of Γ (β);

$$\mathbf{Y}\Gamma_1 + \mathbf{XB}_1 + \mathbf{U}_1 = 0$$

- There are M (in Γ_1) plus K (in B_1) unknown parameters.
- We can normalize one of the Γ_1 , say the coefficient on Y_{i1} , so there are $M + K - 1$ parameters left.
- We can regress Y_{i1} on the K regressors, so we can learn K coefficients that are functions of the $M + K - 1$ unknown parameters, so we are still $M - 1$ restrictions short.

SIMULTANEOUS EQUATIONS MODELS

- Suppose we have some restrictions on Γ_i and B_i that we can write as

$$\mathbf{R}_i \begin{pmatrix} \Gamma_i \\ B_i \end{pmatrix} = 0$$

(these are exclusion restrictions defining implicitly the instruments)

- The we need the number of rows of $\mathbf{R}_i$ to be at least $M - 1$ (order condition).
- We also need the rank of

$$\mathbf{R}_i \begin{pmatrix} \Gamma \\ B \end{pmatrix}$$

to be at least $M - 1$ (rank condition). (this requires that the excluded instruments do affect some of the other endogenous variables.)

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